Awesome Green AI □□

A curated list of awesome Green AI resources and tools to reduce the environmental impacts of using and deploying AI.



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In 2020, Information and Communications Technology (ICT) sector carbon footprint was estimated to be between **2.1-3.9% of total global greenhouse gas emissions**. The ICT sector continues to grow and now dominates other industries (https://www.economist.com/leaders/2017/05/06/the-worlds-most-valuable-resource-is-no-longer-oil-but-data). It is estimated that the **carbon footprint will double to 6-8% by 2025**. For ICT sector to remain compliant with the Paris Agreement, the industry must reduce by 45% its GHG emissions from 2020 to 2030 and reach net zero by 2050 (Freitag et al., 2021 (https://doi.org/10.1016/j.patter.2021.100340)).

AI is one of the fastest growing sectors, disrupting many other industries (AI Market Size Report, 2022 (https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market)). It therefore has an important role to play in reducing carbon footprint. The impacts of ICT, and therefore AI, are **not limited to GHG emissions and electricity consumption**. We need to take into account **all major impacts** (abiotic resource depletion, primary energy consumption, water usage, etc.) using Life Cycle Assessment (LCA) (Arushanyan et al., 2013 (https://doi.org/10.1016/j.compind.2013.10.003)).

AI sobriety not only means optimizing energy consumption and reducing impacts, but also includes studies on **indirect impacts (https://en.wikipedia.org/wiki/Rebound_effect_(conservation)#Direct_and_indirect_effects)** and **rebound effects (https://en.wikipedia.org/wiki/Jevons_paradox)** that can negate all efforts to reduce the environmental footprint (Willenbacher et al. 2021 (https://doi.org/10.1007/978-3-030-88063-7_5)). It is therefore imperative to consider the use of AI before launching a project in order to avoid indirect impacts and rebound effects later on.

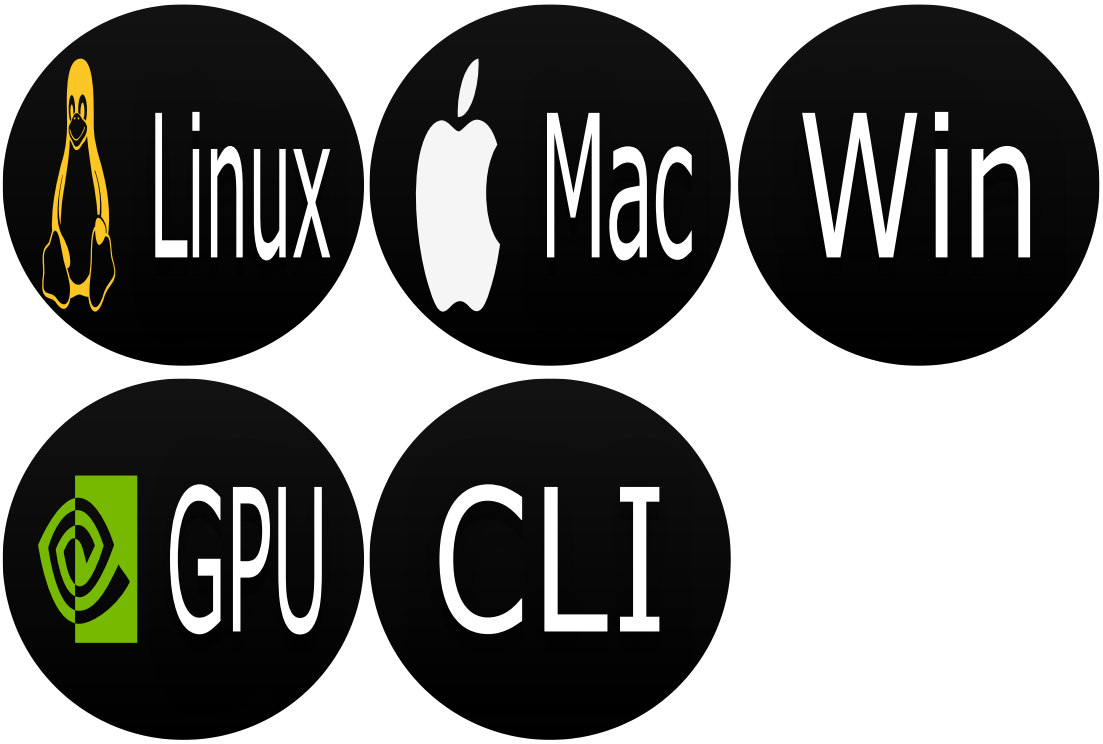
All contributions are welcome. Add links through pull requests (https://github.com/samuelrince/awesome-green-ai/pulls) or create an issue (https://github.com/samuelrince/awesome-green-ai/issues) to start a discussion.

□ Tools

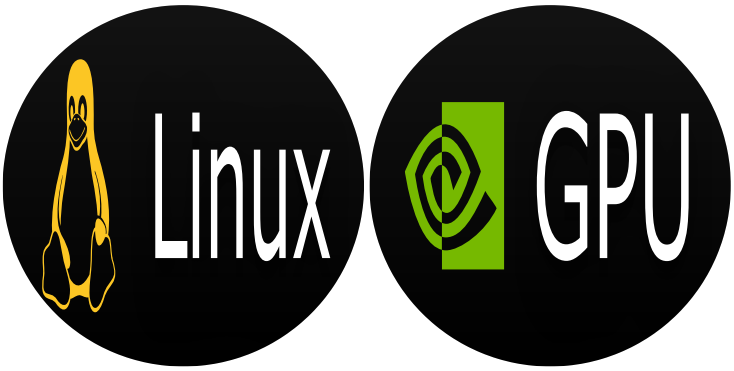
Code-Based Tools

Tools to measure and compute environmental impacts of AI.

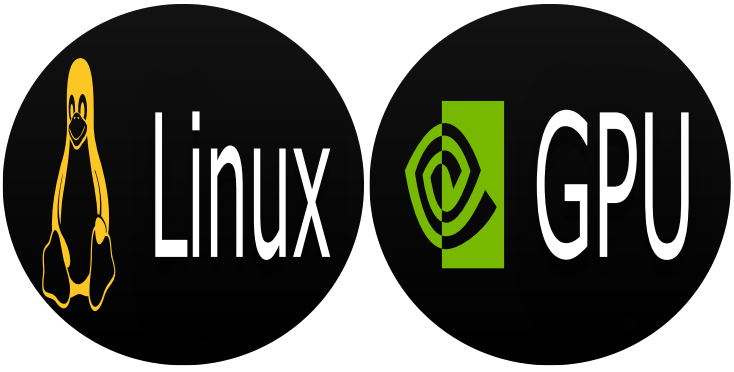
- CodeCarbon (https://github.com/mlco2/codecarbon) – Track emissions from Compute and recommend ways to reduce their impact on the environment.



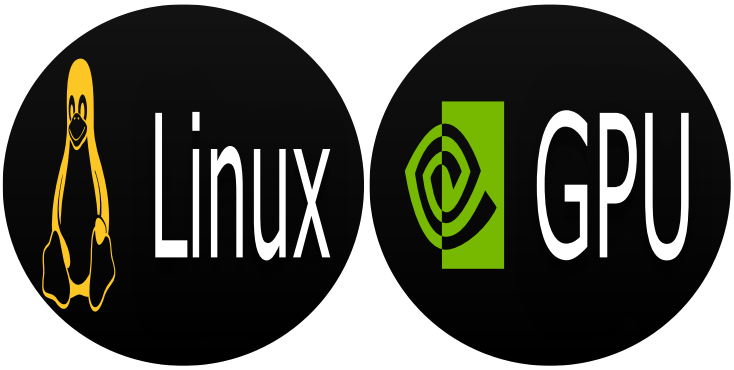
- carbontracker (https://github.com/lfwa/carbontracker) – Track and predict the energy consumption and carbon footprint of training deep learning models.



- Zeus (https://github.com/SymbioticLab/Zeus) – A framework for deep learning energy measurement and optimization.



- Eco2AI (https://github.com/sb-ai-lab/Eco2AI) – A python library which accumulates statistics about power consumption and CO2 emission during running code.



- EcoLogits (https://github.com/genai-impact/ecologits) – Estimates the energy consumption and environmental footprint of LLM inference through APIs.

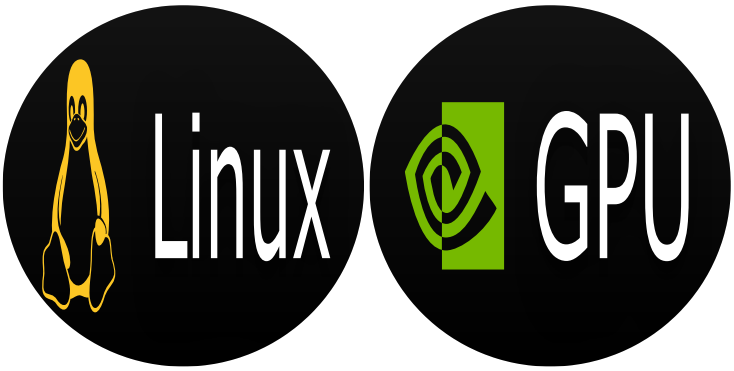




- Tracarbon (https://github.com/fvaleye/tracarbon) – Tracks your device's energy consumption and calculates your carbon emissions using your location.



- AIPowerMeter (https://github.com/GreenAI-Uppa/AIPowerMeter) – Easily monitor energy usage of machine learning programs.

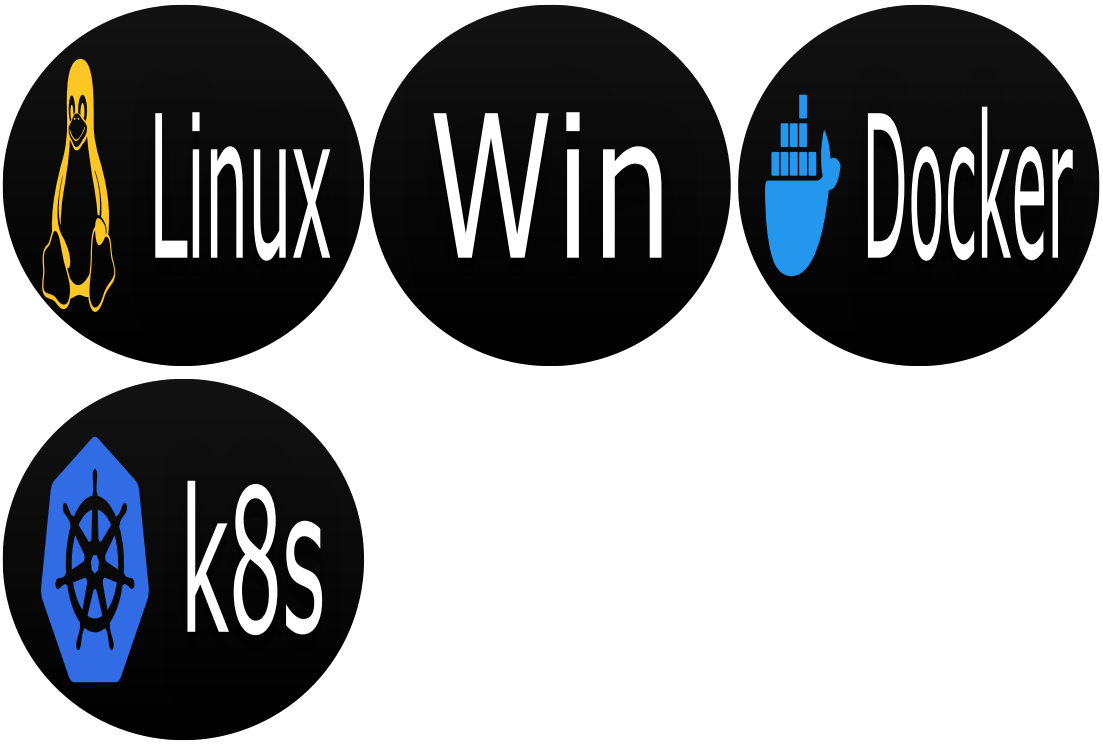


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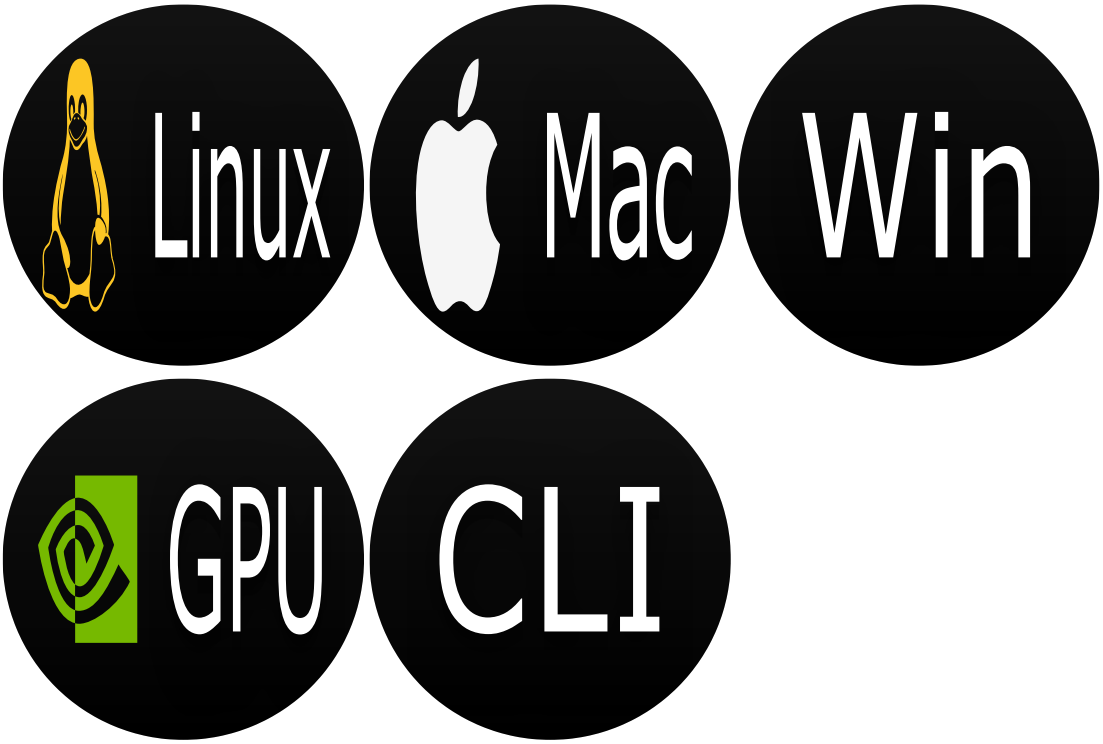
Monitoring Tools

Tools to monitor power consumption and environmental impacts.

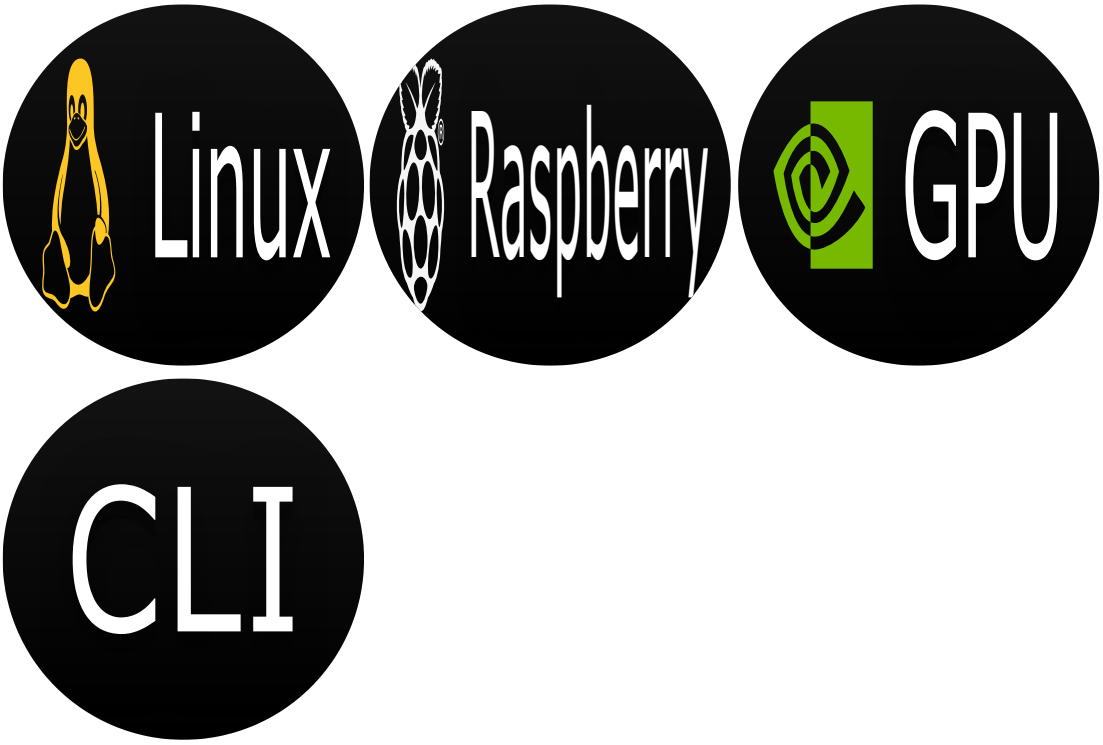
- Scaphandre (https://github.com/hubblo-org/scaphandre) – A metrology agent dedicated to electrical power consumption metrics.



- CodeCarbon (https://github.com/mlco2/codecarbon) – Track emissions from Compute and recommend ways to reduce their impact on the environment.



- PowerJoular (https://github.com/joular/powerjoular) – Monitor power consumption of multiple platforms and processes.



- ALUMET (https://github.com/alumet-dev/alumet) – A modular and efficient software measurement tool.



- cardamon (https://github.com/Root-Branch/cardamon-core) – A tool for measuring the power consumption and carbon footprint of your software.



- Boagent (https://github.com/Boavizta/boagent) – Local API and monitoring agent focussed on environmental impacts of the host.



- Powerletrics (https://github.com/green-kernel/powerletrics) – PowerLetrics is a framework designed to monitor and analyze power consumption metrics at the process level on Linux.

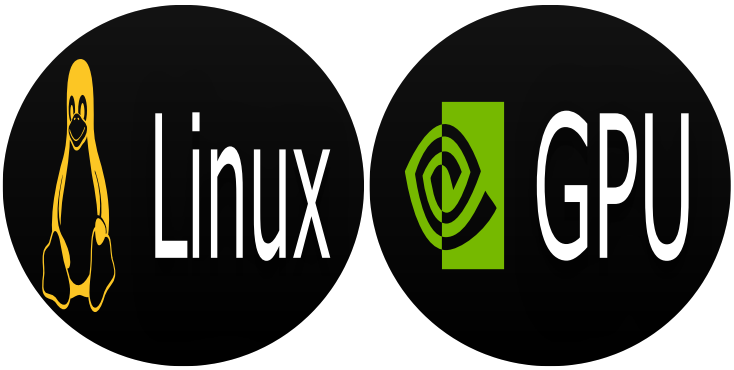


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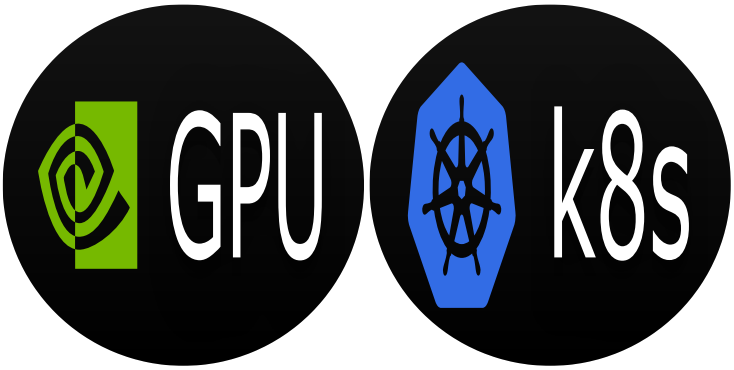
Optimization Tools

Tools to optimize energy consumption or environmental impacts.

- Zeus (https://github.com/SymbioticLab/Zeus) – A framework for deep learning energy measurement and optimization.



- GEOPM (https://github.com/geopm/geopm) – A framework to enable efficient power management and performance optimizations.



Calculation Tools

Tools to estimate environmental impacts of algorithms, models and compute resources.

- Green Algorithms (http://calculator.green-algorithms.org/) - A tool to easily estimate the carbon footprint of a project.
- ML CO2 Impact (https://mlco2.github.io/impact/) - Compute model emissions and add the results to your paper with our generated latex template.
- EcoLogits Calculator (https://huggingface.co/spaces/genai-impact/ecologits-calculator) - Estimate energy consumption and environmental impacts of LLM inference.
- MLCarbon (https://github.com/SotaroKaneda/MLCarbon) - End-to-end carbon footprint modeling tool.
- Carbon footprint modeling tool (https://borisruf.github.io/carbon-footprint-modeling-tool/ai-scenarios.html) - A data model and a viewer for carbon footprint scenarios.
- FLOPs to Footprints (https://huggingface.co/spaces/sophia-falk/flops-2-footprints) - Evaluate the resource cost of AI.

Generic tools:

- Boaviztapi (https://github.com/Boavizta/boaviztapi/) - Multi-criteria impacts of compute resources taking into account manufacturing and usage.
- Datavizta (https://datavizta.boavizta.org/serversimpact) - Compute resources data explorer not limited to AI.
- EcoDiag (https://ecoinfo.cnrs.fr/ecodiag-calcul/) - Compute carbon footprint of IT resources taking into account manufactuing and usage (□□ only).

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Leaderboards

- LLM Perf Leaderboad (https://huggingface.co/spaces/optimum/llm-perf-leaderboard) - Benchmarking LLMs on performance and energy.
- ML.Energy Leaderboard (https://ml.energy/leaderboard/?__theme=light) - Energy consumption of GenAI models at inference.
- AI Energy Score Leaderboard (https://huggingface.co/spaces/AIEnergyScore/2024_Leaderboard) - Energy efficiency ratings for AI models.

Papers

- Energy and Policy Considerations for Deep Learning in NLP - Strubell et al. (2019) (https://arxiv.org/abs/1906.02243)
- Quantifying the Carbon Emissions of Machine Learning - Lacoste et al. (2019) (https://arxiv.org/abs/1910.09700)
- Carbontracker: Tracking and Predicting the Carbon Footprint of Training Deep Learning Models - Anthony et al. (2020) (https://arxiv.org/abs/2007.03051)
- The carbon impact of artificial intelligence. - Payal Dhar (2020) (https://www.researchgate.net/profile/Payal-Dhar-2/publication/343618995_The_carbon_impact_of_artificial_intelligence/links/63051d3a5eed5e4bd114250a/The-carbon-impact-of-artificial-intelligence.pdf)
- Green AI - Schwartz et al. (2020) (https://cacm.acm.org/magazines/2020/12/248800-green-ai/fulltext)
- Towards the Systematic Reporting of the Energy and Carbon Footprints of Machine Learning - Henderson et al. (2020) (https://www.jmlr.org/papers/v21/20-312.html)
- GPU Lifetimes on Titan Supercomputer: Survival Analysis and Reliability - Ostrouchov et al. (2020) (https://ieeexplore.ieee.org/document/9355319)
- The Energy and Carbon Footprint of Training End-to-End Speech Recognizers - Parcollet et al. (2021) (https://hal.science/hal-03190119v1/)
- Carbon Emissions and Large Neural Network Training - Patterson, et al. (2021) (https://arxiv.org/abs/2104.10350)
- Green Algorithms: Quantifying the Carbon Footprint of Computation - Lannelongue et al. (2021) (https://onlinelibrary.wiley.com/doi/10.1002/advs.202100707)
- Aligning artificial intelligence with climate change mitigation - Kaack et al. (2021) (https://hal.archives-ouvertes.fr/hal-03368037/document)
- A Practical Guide to Quantifying Carbon Emissions for Machine Learning researchers and practitioners - Ligozat et al. (2021) (https://hal.archives-ouvertes.fr/hal-03376391/document)
- Unraveling the Hidden Environmental Impacts of AI Solutions for Environment Life Cycle Assessment of AI Solutions - Ligozat et al. (2022) (https://arxiv.org/abs/2110.11822)
- Measuring the Carbon Intensity of AI in Cloud Instances - Dodge et al. (2022) (https://arxiv.org/abs/2206.05229)
- Green AI: do deep learning frameworks have different costs? - Georgiou et al. (2022) (https://discovery.ucl.ac.uk/id/eprint/10143851/1/ml_performance_study_preprint.pdf)
- Estimating the Carbon Footprint of BLOOM a 176B Parameter Language Model - Luccioni et al. (2022) (https://arxiv.org/abs/2211.02001)
- Bridging Fairness and Environmental Sustainability in Natural Language Processing - Hessenthaler et al. (2022) (https://arxiv.org/abs/2211.04256)
- Eco2AI: carbon emissions tracking of machine learning models as the first step towards sustainable AI - Budennyy et al. (2022) (https://arxiv.org/abs/2208.00406)
- Environmental assessment of projects involving AI methods - Lefèvre et al. (2022) (https://hal.science/hal-03922093)
- Sustainable AI: Environmental Implications, Challenges and Opportunities - Wu et al. (2022) (https://arxiv.org/abs/2111.00364)
- The Carbon Footprint of Machine Learning Training Will Plateau, Then Shrink - Patterson et al. (2022) (https://arxiv.org/abs/2204.05149)
- Towards the Systematic Reporting of the Energy and Carbon Footprints of Machine Learning - Henderson et al. (2022) (https://arxiv.org/abs/2002.05651)
- Towards Sustainable Artificial Intelligence: An Overview of Environmental Protection Uses and Issues - Pachot et al. (2022) (https://arxiv.org/abs/2212.11738)

- Method and evaluations of the effective gain of artificial intelligence models for reducing CO2 emissions - Delanoë et al. (2023) (https://www.sciencedirect.com/science/article/pii/S030147972300049X)
- Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models - Li et al. (2023) (https://arxiv.org/abs/2304.03271)
- Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training - You et al. (2023) (https://www.usenix.org/conference/nsdi23/presentation/you)
- Trends in AI inference energy consumption: Beyond the performance-vs-parameter laws of deep learning Desislavov et al. (2023) (https://www.sciencedirect.com/science/article/pii/S2210537923000124)
- Chasing Low-Carbon Electricity for Practical and Sustainable DNN Training - Yang et al. (2023) (https://www.climatechange.ai/papers/iclr2023/29)
- Toward Sustainable HPC: Carbon Footprint Estimation and Environmental Implications of HPC Systems - Li et al. (2023) (https://arxiv.org/abs/2306.13177)
- Reducing the Carbon Impact of Generative AI Inference (today and in 2035) - Chien et al. (2023) (https://dl.acm.org/doi/10.1145/3604930.3605705)
- LLMCarbon: Modeling the End-To-End Carbon Footprint of Large Language Models - Faiz et al. (2023) (https://arxiv.org/abs/2309.14393)
- The growing energy footprint of artificial intelligence - De Vries (2023) (https://www.sciencedirect.com/science/article/pii/S2542435123003653)
- Exploring the Carbon Footprint of Hugging Face's ML Models: A Repository Mining Study - Castano et al. (2023) (https://ieeexplore.ieee.org/document/10304801)
- Exploding AI Power Use: an Opportunity to Rethink Grid Planning and Management - Lin et al. (2023) (https://arxiv.org/abs/2301.03148)
- Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models - Li et al. (2023) (https://arxiv.org/abs/2304.03271)
- Power Hungry Processing: Watts Driving the Cost of AI Deployment? - Luccioni et al. (2023) (https://arxiv.org/abs/2311.16863)
- Perseus: Removing Energy Bloat from Large Model Training - Chung et al. (2023) (https://arxiv.org/abs/2312.06902)
- Timeshifting strategies for carbon-efficient long-running large language model training - Jagannadharao et al. (2023) (https://link.springer.com/article/10.1007/s11334-023-00546-x)
- From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference - Samsi et al. (2023) (https://arxiv.org/pdf/2310.03003)
- Estimating the environmental impact of Generative-AI services using an LCA-based methodology - Berthelot et al. (2024) (https://hal.univ-lorraine.fr/INRIA/hal-04346102v2)
- Towards Greener LLMs: Bringing Energy-Efficiency to the Forefront of LLM Inference - Stojkovic et al. (2024) (https://arxiv.org/abs/2403.20306)
- Green AI: Exploring Carbon Footprints, Mitigation Strategies, and Trade Offs in Large Language Model Training - Liu et al. (2024) (https://arxiv.org/abs/2404.01157)
- Engineering Carbon Emission-aware Machine Learning Pipelines - Humsom et al. (2024) (https://dl.acm.org/doi/10.1145/3644815.3644943)
- A simplified machine learning product carbon footprint evaluation tool - Lang et al. (2024) (https://www.sciencedirect.com/science/article/pii/S2666789424000254)
- Measuring and Improving the Energy Efficiency of Large Language Models Inference - Argerich et al. (2024) (https://ieeexplore.ieee.org/document/10549890)
- Beyond Efficiency: Scaling AI Sustainably - Wu et al. (2024) (https://arxiv.org/abs/2406.05303)
- The Price of Prompting: Profiling Energy Use in Large Language Models Inference - Huson et al. (2024) (https://arxiv.org/abs/2407.16893)
- Offline Energy-Optimal LLM Serving: Workload-Based Energy Models for LLM Inference on Heterogeneous Systems - Wilkins et al. (2024) (https://arxiv.org/abs/2407.04014)
- MLCA: a tool for Machine Learning Life Cycle Assessment - Morand et al. (2024) (https://hal.science/hal-04643414)

- Hype, Sustainability, and the Price of the Bigger-is-Better Paradigm in AI - Varoquaux et al. (2024) (https://arxiv.org/abs/2409.14160)
- Addition is All You Need for Energy-efficient Language Models - Luo et al. (2024) (https://arxiv.org/abs/2410.00907)
- E-waste challenges of generative artificial intelligence - Wang et al. (2024) (https://www.nature.com/articles/s43588-024-00712-6)
- Green My LLM: Studying the key factors affecting the energy consumption of code assistants - Coignion et al. (2024) (https://arxiv.org/abs/2411.11892)
- LLM-Inference-Bench: Inference Benchmarking of Large Language Models on AI Accelerators - Chitty-Venkata et al. (2024) (https://arxiv.org/abs/2411.00136)
- A Beginner's Guide to Power and Energy Measurement and Estimation for Computing and Machine Learning - Jagannadharao et al. (2024) (https://arxiv.org/abs/2412.17830v1)
- From Efficiency Gains to Rebound Effects: The Problem of Jevons' Paradox in AI's Polarized Environmental Debate - Luccioni et al. (2025) (https://arxiv.org/abs/2501.16548)
- Understanding the environmental impact of generative AI services - Berthelot et al. (2025) (https://hal.science/hal-04920612)
- EcoServe: Designing Carbon-Aware AI Inference Systems - Li et al. (2025) (https://arxiv.org/abs/2502.05043)
- Towards Sustainable NLP: Insights from Benchmarking Inference Energy in Large Language Models - Poddar et al. (2025) (https://arxiv.org/abs/2502.05610)
- Unveiling Environmental Impacts of Large Language Model Serving: A Functional Unit View - Wu et al. (2025) (https://arxiv.org/abs/2502.11256)
- Beyond Test-Time Compute Strategies: Advocating Energy-per-Token in LLM Inference - Wilhelm et al. (2025) (https://dl.acm.org/doi/10.1145/3721146.3721953)
- Frugal AI: Introduction, Concepts, Development and Open Questions - Arga et al. (2025) (https://www.researchgate.net/publication/390920260_Frugal_AI_Introduction_Concepts_Development_and_Open_Questions)
- Energy Considerations of Large Language Model Inference and Efficiency Optimizations - Fernandez et al. (2025) (https://arxiv.org/abs/2504.17674)
- The ML.ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization - Chung et al. (2025) (https://arxiv.org/abs/2505.06371)
- How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference - Jegham et al. (2025) (https://arxiv.org/abs/2505.09598)
- Breaking the ICE: Exploring promises and challenges of benchmarks for Inference Carbon & Energy estimation for LLMs - Sikand et al. (2025) (https://arxiv.org/abs/2506.08727)
- Not All Water Consumption Is Equal: A Water Stress Weighted Metric for Sustainable Computing - Wu et al. (2025) (https://arxiv.org/abs/2506.22773)
- Measuring the environmental impact of delivering AI at Google Scale - Elsworth et al. (2025) (https://services.google.com/fh/files/misc/measuring_the_environmental_impact_of_delivering_ai_at_google_scale.pdf)
- More than Carbon: Cradle-to-Grave environmental impacts of GenAI training on the Nvidia A100 GPU - Falk et al. (2025) (https://arxiv.org/abs/2509.00093) [supplemental material (https://github.com/sophia-falk/more-than-carbon)]
- Video Killed the Energy Budget: Characterizing the Latency and Power Regimes of Open Text-to-Video Models - Delavande et al. (2025) (https://arxiv.org/abs/2509.19222)
- Ground-Truthing AI Energy Consumption: Validating CodeCarbon Against External Measurements - Fischer (2025) (https://arxiv.org/abs/2509.22092)
- Green Prompt Engineering: Investigating the Energy Impact of Prompt Design in Software Engineering - De Martino et al. (2025) (https://arxiv.org/abs/2509.22320)
- From FLOPs to Footprints: The Resource Cost of Artificial Intelligence - Falk et al. (2025) (https://arxiv.org/abs/2512.04142)
- Beyond Counting Carbon: AI Environmental Assessments Struggle to Inform Net Impact Decisions - Cook et al. (2025) (https://doi.org/10.3929/ethz-c-000789254)
- Kareus: Joint Reduction of Dynamic and Static Energy in Large Model Training - Wu et al. (2026) (https://arxiv.org/abs/2601.17654)

- Where Do the Joules Go? Diagnosing Inference Energy Consumption - Chung et al. (2026) (https://arxiv.org/abs/2601.22076)
- Small Talk, Big Impact: The Energy Cost of Thanking AI - Delavande et al. (2026) (https://arxiv.org/abs/2601.22357)
- Understanding Efficiency: Quantization, Batching, and Serving Strategies in LLM Energy Use - Delavande et al. (2026) (https://arxiv.org/abs/2601.22362)
- From Attributional to Consequential LCA: Which Theoretical Framework for Assessing AI's Environmental Impacts? - Ekchajzer et al. (2026) (https://hal.science/hal-05512793/)
- Small Bottle, Big Pipe: Quantifying and Addressing the Impact of Data Centers on Public Water Systems - Han et al. (2026) (https://arxiv.org/abs/2603.02705)
- Green Prompting: Characterizing Prompt-driven Energy Costs of LLM Inference - Adamska et al. (2026) (https://arxiv.org/abs/2503.10666)
- The Energy Footprint of LLM-Based Environmental Analysis: LLMs and Domain Products - Bao et al. (2026) (https://arxiv.org/abs/2604.00053)

Survey Papers

- Evaluating the carbon footprint of NLP methods: a survey and analysis of existing tools - Bannour et al.(2021) (https://aclanthology.org/2021.sustainlp-1.2.pdf)
- A Survey on Green Deep Learning - Xu et al. (2021) (https://arxiv.org/abs/2111.05193)
- A Systematic Review of Green AI - Verdecchia et al. (2023) (https://arxiv.org/abs/2301.11047)
- Counting Carbon: A Survey of Factors Influencing the Emissions of Machine Learning - Luccioni et al. (2023) (https://arxiv.org/abs/2302.08476)
- How to estimate carbon footprint when training deep learning models? A guide and review - Bouza et al. (2023) (https://iopscience.iop.org/article/10.1088/2515-7620/acf81b/meta)
- Towards Efficient Generative Large Language Model Serving: A Survey from Algorithms to Systems - Miao et al. (2023) (https://arxiv.org/abs/2312.15234)

□ Reports

- The great challenges of generative AI (□□ only) - Data For Good 2023 (https://s3.fr-par.scw.cloud/strapi-uploads/Data_For_Good_Livre_Blanc_IA_Generative_v1_0_1_345b2a7454.pdf)
- General framework for frugal AI - AFNOR 2024 (https://www.boutique.afnor.org/en-gb/standard/afnor-spec-2314//fa208976/421140)
- Powering Up Europe: AI Datacenters and Electrification to Drive +c.40%-50% Growth in Electricity Consumption - Goldman Sachs 2024 (https://www.goldmansachs.com/insights/goldman-sachs-research/electrify-now-powering-up-europe)
- Generational Growth — AI/data centers' global power surge and the sustainability impact - Goldman Sachs 2024 (https://www.goldmansachs.com/insights/goldman-sachs-research/gs-sustain-generational-growth-ai-data-centers-global-power)
- AI and the Environment - International Standards for AI and the Environment - ITU 2024 (https://www.itu.int/dms_pub/itu-t/opb/env/T-ENV-ENV-2024-1-PDF-E.pdf)
- Powering artificial intelligence: a study of AI's footprint—today and tomorrow - Deloitte 2024 (https://www.deloitte.com/global/en/issues/climate/powering-ai.html)
- Artificial Intelligence and Electricity: A System Dynamics Approach - Schneider Electric 2024 (https://www.se.com/ww/en/insights/sustainability/sustainability-research-institute/artificial-intelligence-electricity-system-dynamics-approach/)
- Developing sustainable Gen AI - Capgemini 2025 (https://www.capgemini.com/gb-en/insights/research-library/sustainable-gen-ai/)
- Exploring the sustainable scaling of AI dilemma: A projective study of corporations' AI environmental impacts - Capgemini Invent 2025 (https://arxiv.org/abs/2501.14334)

- Intelligence artificielle, données, calculs : quelles infrastructures dans un monde décarboné (□□ only) - The Shift Project 2025 (https://theshiftproject.org/article/rapport-intermediaire-ia/)
- Measuring the environmental impacts of artificial intelligence compute and applications OECD 2025 (https://www.oecd.org/content/dam/oecd/en/publications/reports/2022/11/measuring-the-environmental-impacts-of-artificial-intelligence-compute-and-applications_3dddded5/7babf571-en.pdf)
- Recommendation ITU-T L.1801 - Guidelines for assessing the environmental impact of artificial intelligence systems ITU 2026 (https://www.itu.int/epublications/publication/itu-t-l-1801-2026-02-guidelines-for-assessing-the-environmental-impact-of-artificial-intelligence-systems)